

Cable TV technology for local access

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Cable TV networks will pass 17 million homes in the UK by early next century. Reliability has improved dramatically in recent years due to the widespread introduction of fibre into the network. Possibly the biggest attraction of cable is the enormous bandwidth that is available, together with its high degree of flexibility. This flexibility can be utilised to accommodate new services such as digital TV, data and telephony.

1. Introduction

By early next century, cable TV will be available to 17 million homes in the UK.

Using modern technology, cable TV networks can simultaneously deliver hundreds of digital TV channels, tens of analogue TV channels, high-speed data and telephony, to tens of thousands of homes in a cable franchise area of perhaps 125 000 homes. Frequency division multiplexing allows operators to mix many different services on the same transmission medium with virtually no unwanted interaction between the services. Because there is no unwanted interaction, individual services can be added and removed with little effect on customers taking other services.

Cable TV technology has evolved over the years to allow cable to become a highly sophisticated broadband transmission medium, capable of meeting the demands of customers and service providers alike. Cable technology has not stopped evolving, and the promise of more digital bandwidth to every customer can be met with more sophisticated forms of modulation. By utilising the latest advances in digital signal processing, more advanced and robust transmission protocols can be used, increasing the penetration of fibre into the access network. Already there are some networks where fibre reaches to within a few hundred metres of each home. Could cable become the first access network to offer fibre to the home?

2. Overview of modern cable TV systems

In much of Europe broadcast TV signals are encoded as phase alternation line (PAL) for transmission. The PAL signal is then modulated on to a radio frequency carrier using amplitude modulated vestigial sideband (AM-VSB) by the broadcasters to enable the TV signal to be transmitted at radio frequencies. Reception of the broadcast-modulated TV signals requires an exceptionally high carrier-to-

noise ratio, if a high-quality TV picture is to be received by the viewer. Early cable TV systems were introduced to ensure adequate reception of the TV signals in areas where natural obstructions such as mountains and high hills blocked the direct path of the radio signal. By placing a suitable aerial on a high point, where the usable signals could be received, it was possible to then send the signals down a length of co-axial cable to the nearby community and directly into the viewer's TV set, thus ensuring adequate TV reception. If the coaxial cable run was too long, suitable coaxial amplifiers were connected along the cable to maintain signal level. These early systems were often privately owned and operated.

Early commercial systems were also designed around an amplified coaxial network (tree and branch). The 1970s systems had VHF (300 MHz) trunks with local conversion of up to 7 channels to UHF (860 MHz) for the distribution network. Subsequently a few extra premium video channels were added. In the 1980s systems were still VHF but featured a bandwidth of 450 MHz all the way to the customer and could deliver up to 30 video channels. A return path spectrum was also included but rarely used for service applications.

In the early 1980s BT designed an advanced switched star network [1] which brought fibre to the street cabinet and a coaxial final drop to the customer. It was designed to carry 24 video channels with an interactive capability. This interactivity allowed a full video-on-demand system to be operated, together with home shopping. The optical network used multimode fibre and 850 nm lasers to deliver multichannel-per-fibre TV from the head-end to the cabinet and control data in both directions of transmission. The coaxial drop to the customer could be up to 500 m long because the system used a switch within the cabinet and delivered only the selected video channels or text and 12

FM radio channels to the customer thus keeping the bandwidth to a minimum.

In tree-and-branch topologies used in all other operations, long cascades of amplifiers proved difficult to maintain, and the reliability and the signal quality of the systems suffered. To overcome this problem cable networks were upgraded in the early 1990s to incorporate fibre feeds to local areas (typically 1500 to 2500 homes) and higher bandwidth amplifiers and taps (600 MHz). This eliminated the long cascades of amplifiers. Typically the maximum cascade of amplifiers was reduced to 9 (6 trunk and 3 distribution), against a worst case, prior to this, of more than 20. The fibre systems were 1310 nm single mode with high reliability electro-optical components.

These changes greatly improved the performance of the systems and increased channel capacities to around 40 to 45. These networks are, today, commonly known as hybrid fibre/coaxial (HFC). Return paths (upstream) were provided but rarely used.

Today, new networks are being constructed with fibre to a single distribution launch amplifier (DLA), thus reducing the optical node size to 500 homes or less. The optical equipment, amplifiers and passives (taps and coaxial splitters) are often specified to 860 MHz; this allows a theoretical analogue channel capacity of more than 80 channels. However, avoiding frequencies which are already used by radio devices, and keeping the loading on the amplifiers to within their optimum performance areas, reduces this to around 60 PAL-I video channels.

2.1 Hybrid/fibre co-axial topology

There are many different fibre-based cable TV topologies available to the cable planner and almost all of

these can be called HFC. In general, where fibre feeders are used to deliver a multiplex of TV channels deep into the network, with subsequent delivery over coaxial cable, this can be described as an HFC network.

The advantage of using fibre in the network is because of the very low transmission loss of this medium compared with coaxial cable. Typically a fibre link will have an installed loss of around 0.4 dB/km compared with 60 dB/km for coaxial trunk cable. In practice the coaxial cable would need three amplifiers to span a single kilometre, and as many as 20 or 30 where splitting and taps are used. A single fibre link can span as much as 50 km between optical transmitter and the fibre serving area.

On its own, fibre has little advantage without the availability of low distortion/low noise lasers and suitable optical receivers to transmit and receive the multiplex of channels to be carried. In order to achieve the very high signal-to-noise ratios required for the successful transmission of the multichannel TV multiplex, CATV optical links are operated at relatively high optical levels compared to typical digital transmission systems. Optical receive levels need to be kept high and this leads to very small optical loss budgets, typically 10 to 13 dB. Optical splitting is therefore rarely used and individual fibres are used to reach each serving area in the network.

Figure 1 shows how separate fibres carry the signals to the different optical nodes from where signals are distributed over coaxial cable to the customers' premises. Depending on the size of franchise and its location, as many as 100 fibre feeds may be required. Return path requires a separate fibre from each optical node to the headend. At low penetration rates this architecture can be expensive to build but it does provide excellent signal quality and reliability due to the reduced number of coaxial amplifiers used.

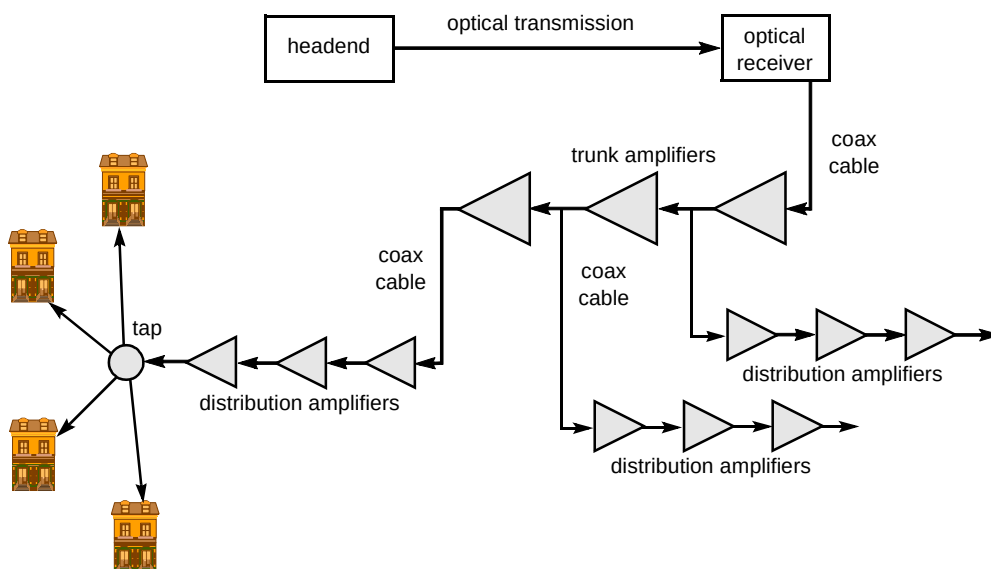


Fig 1 The basic HFC architecture.

Examples of variations on this architecture include fibre backbone, forward intermediate/terminating trunk (FITT), Scientific Atlanta's Fibre to the Serving Area (FSA), Phillips' Diamond network, Jerrold's Starburst network and CableLabs' structured network. Each has some advantage in particular situations and planners need to consider which best suits their need for the areas to be covered.

2.2 HFC technology

This section considers the principal technologies used to distribute cable TV signals across an HFC network.

Set-top box

Early cable television systems did not require the use of any additional equipment as the signals transmitted over the cable network were in the same frequency band as that of conventional off-air signals. Consequently, it was possible to resolve these signals using a standard domestic television tuner. With the increase in channel availability, a method had to be found of transporting these additional channels. This new method would also need to include a suitable form of access control that would only allow customers to view the channels to which they subscribed.

Unfortunately these additional requirements were outside the capability of a normal domestic television receiver, and as a consequence additional customer equipment was required. This additional equipment took the form of a set-top unit located in close proximity to the customer's standard domestic TV. The principal purpose of the early analogue set-top units was to frequency translate and provide access control to differing service packages.

With advances in technology, the set-top unit is now seen as a critical piece of equipment in the customer's premises, giving additional facilities to the customer such as impulse pay per view and video on demand.

With the recent advances in digital video broadcasting and digital TV, the set-top box is now evolving to be a very sophisticated piece of equipment with a computing power similar to that of a high specification PC. This is required to support full MPEG2 picture decoding and sound, together with electronic program guides, Web browsing, and some very sophisticated access network control.

Coaxial amplifiers

As the demand for bandwidth increases, coaxial amplifiers increasingly become key network components. Improvements in technology have led to amplifiers capable of bandwidths of at least 50 to 860 MHz (the highest frequency in the UHF broadcast band) and with distortion low enough to allow at least 60 PAL-I channels to be transmitted (PAL-I is the UK TV transmission standard). Return path amplifier modules usually operate from 5 MHz to between 30 and 200 MHz, depending on the split chosen for the network. Figures 2a and 2b show a typical amplifier.

Coaxial amplifiers are needed in cable TV networks to compensate for the loss incurred by the network coaxial cable and to ensure an adequate level to each customer premises from the final tap. They are usually powered over the coaxial cable from a convenient power insertion point and can in turn act as power bypasses to pass power to the next amplifier in the cascade.

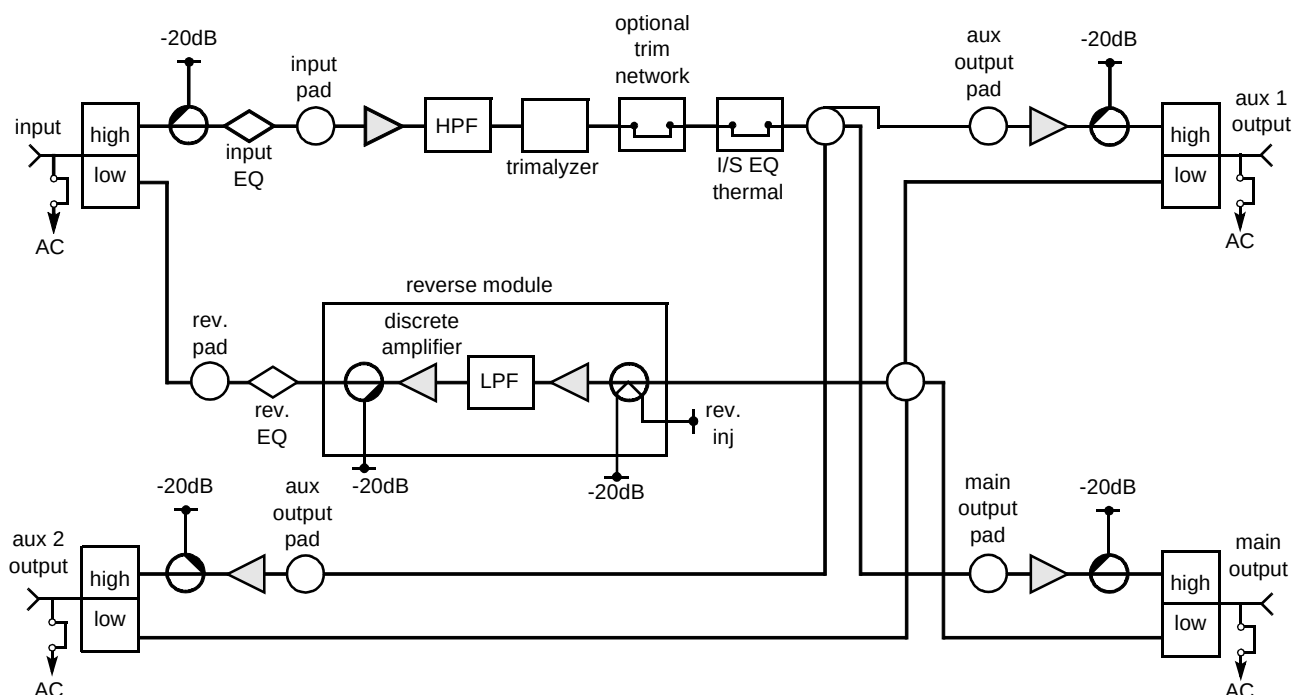


Fig 2a Block diagram of a typical coaxial amplifier with reverse path module installed (courtesy of Scientific Atlanta).

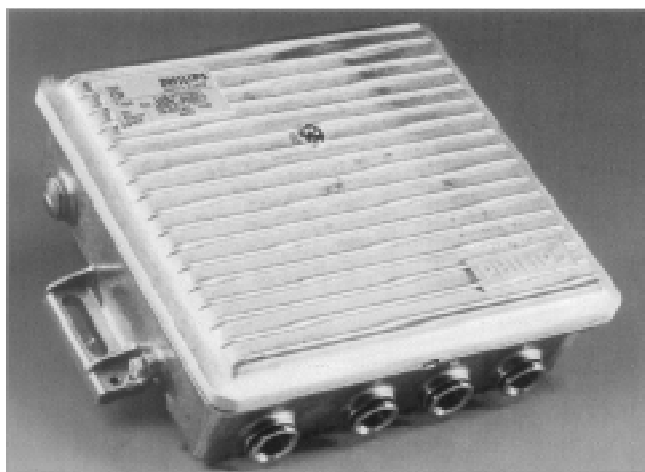


Fig 2b Typical coaxial amplifier in its aluminium housing (courtesy of Philips Broadband).

Typically, an amplifier consists of an outer casing or housing in which is mounted the main amplifier together with the diplexer, equaliser, attenuator, AGC control, monitor, power conditioner and return amplifier module. Depending on its position in the network the amplifier may also house a splitter to give multiple outputs.

Amplifiers are not usually designed for full immersion in water, hence they are normally housed above ground in distinctive 'cable' cabinets similar in size and shape to conventional telephone company distribution cabinets. The availability of waterproof sleeves suitable for housing certain types of amplifier has allowed some cable companies to mount their amplifiers within footway boxes; however, this is the exception rather than the rule.

Amplifiers are usually designed to operate with local or coaxial line powering from between 45 and 90 V AC. This must be converted to DC to power the various stages which usually require 24 V at several amps. This is achieved with the power conditioning circuit.

In order to achieve high efficiency, and hence reduce unwanted heating as well as power wastage, switching regulators are often used, with efficiencies as high as 90%. Power conditioning usually also contains fusing, surge arrest and test points as well as the regulator.

Optical transmitters

Probably the single biggest development in cable TV technology in the last 15 years has been the introduction of optical transmission. Prior to the mid 1980s optical fibre was already being used to transmit multichannel TV between cable headends and hub sites. However, these fibre links tended to use frequency modulation (FM) rather than native AM-VSB as used elsewhere in the network for transmission. Consequently costly modulation mode

conversion was required to and from these links. FM links of this type are still in use in many places today.

The breakthrough came with the development of semiconductor lasers with sufficiently low relative intensity noise (RIN) to allow the required high carrier-to-noise ratios (CNRs) required by PAL and NTSC encoded AM-VSB transmission. Early lasers used to transmit AM-VSB carriers were limited by intermodulation distortion to just a few channels. Further development of distributed feedback (DFB) lasers together with predistortion circuits resulted in exceptionally linear performance, while maintaining the required RIN. This permitted optical transmission of up to 100 (NTSC) TV channels over distances of 30 to 50 km.

Externally modulated optical transmitters using either DFB lasers or YAG (yttrium aluminium garnet) sources have further improved performance, but generally at higher cost than the directly modulated laser transmitters.

Optical transmitter output power tends to be in the range +7 to +20 dBm.

Optical receivers

In order to maintain CNR performance CATV optical receivers have to be operated well above the thermal noise floor of the receiver. This usually means operating with receiver optical input levels of 0 to -6 dBm. At these levels optical detector shot noise is greater than thermal noise and would normally be expected to dominate the noise performance of the optical link. However, shot noise is proportional to optical power, whereas the laser noise is proportional to the square of the optical power. As the received optical power increases, the laser noise dominates and determines the available CNR.

Optical links are usually designed to maximise optical budget. A well-designed link will operate where the receiver shot noise just starts to dominate overall link noise. This is a level well above that normally expected with other optical transmission systems and can lead to severe distortion of the received signal (intermodulation). Manufacturers have developed matching techniques where the receiver can operate at these high optical levels reliably, with minimum distortion and low noise.

Fibre amplifiers

Currently, most optical links operate at 1310 nm. Some applications demand more optical performance than the laser transmitter and optical receiver combination can provide. In these cases erbium-doped fibre amplifiers (EDFAs) can be used to give several high-level outputs from one modulated source. Operating at 1550 nm, EDFAs require that the optical link also operates at 1550 nm. For this reason EDFAs are most often used on long trunk routes

where the lower loss of 1550 nm fibre can also help increase reach significantly.

The optical link

Depending on the architecture of the network, optical transmitters are situated at the cable headend and feed up to 50 km of optical fibre. Optical receivers are often accommodated in the same housing as an amplifier. It is quite common to house the receiver in the lid of the amplifier with the accompanying coaxial amplifier in the base of the housing, forming a self-contained optical node.

With this type of arrangement a return fibre is used for upstream. Typical downstream lasers would be too expensive to use for the limited bandwidth needed in the upstream direction and the number of channels likely to be needed is small. For this reason upstream lasers to drive the upstream fibre are often low-cost Fabry Perot (FP). Although the upstream bandwidth may be no more than 5—50 MHz, advantage can be taken of the greater available bandwidth of the FP laser and up to four bands of 5 to 50 MHz can be frequency division multiplexed together to occupy 5—200 MHz and still retain adequate upstream performance.

Just as a low-cost laser transmitter is required at each optical node, so a complementary optical receiver is required at the headend for each fibre node. These are often similar to the downstream optical receivers, but specified to only 200 MHz. As with optical receivers, upstream laser transmitters are often housed in the same optical node casing. Figure 3 shows a typical CATV optical transmitter.

3. Cable spectrum

Cable operators can use any frequency plan they choose, although most countries do impose some regulations about frequencies (channels) that may not be used in cable systems because of potential interference to other radio services using the same frequency range. This is particularly true in the case of aircraft emergency frequencies.

Cable channels are spaced according to the prevailing broadcast channel spacing in that country. In North America channels are spaced at 6 MHz because NTSC transmissions are designed to work with that spacing. In Europe a

combination of 7 and 8 MHz is used to allow for the various PAL and Secam encoding requirements.

Figure 4 shows a technician measuring signal level, while Fig 5 shows a typical frequency plan that might be used in the UK. Unavailable channels are not shown in Fig 5. The higher channels are usually allocated to digital services on the basis that the lower CNR available at this end of the spectrum can still be used for digital transmission. In practice, digital channels might initially be accommodated in unused gaps between analogue channels to take advantage of a digital transmission's ability to operate more reliably in the presence of low-level interference.



Fig 4 Cable technician measuring signal level at a street cabinet (courtesy of Philips Broadband).

Upstream bandwidth extends from 5—50 MHz. The lower frequencies are more prone to ingress from powerful short-wave broadcast stations and radio amateur transmissions and are therefore often avoided by cable companies who prefer to use the more limited band from about 20 to 50 MHz.

4. Cable TV services

4.1 Analogue TV

A PAL-I baseband video signal is considered to have a bandwidth from DC to 5.5 MHz for the luminance and chrominance portion of the signal. The sound is carried sep-

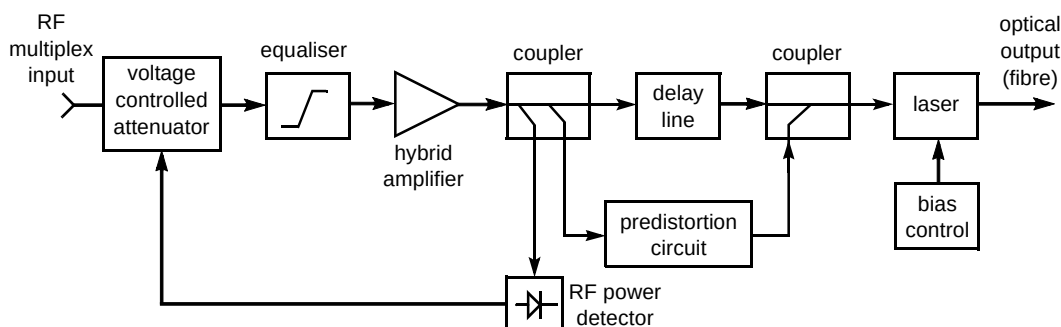


Fig 3 A CATV optical transmitter.

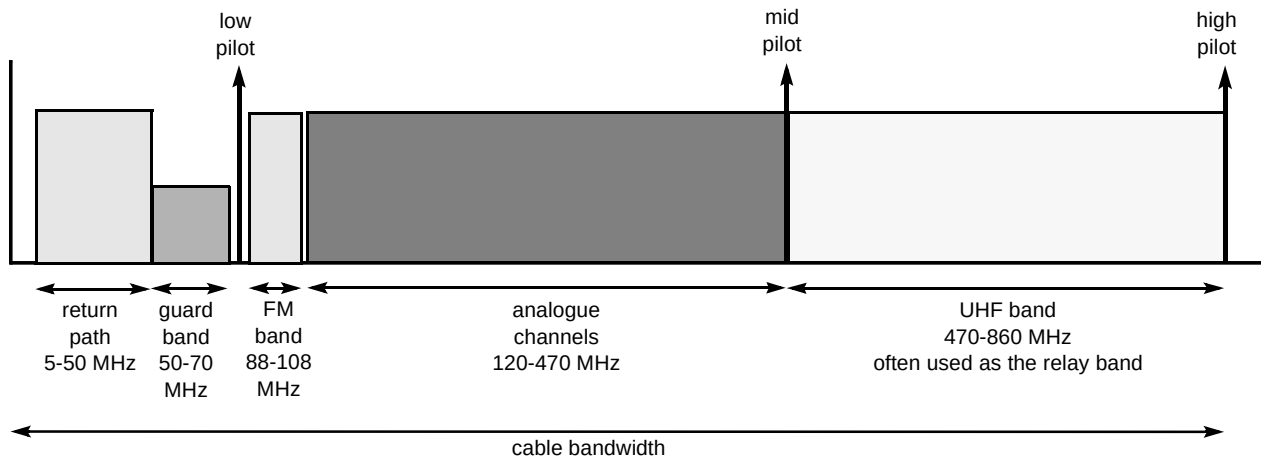


Fig 5 A typical UK cable frequency plan.

arately on a frequency modulated (FM) 6-MHz subcarrier for the mono sound and QPSK modulated on to a 6.552 MHz subcarrier for digital NICAM stereo. The addition of these sound subcarriers gives an overall bandwidth requirement from about DC—6.9 MHz.

Within the UK the channel spacing for terrestrial and cable transmission is divided into 8 MHz slots. Each of these slots has sufficient bandwidth to accommodate a single PAL-I signal following vestigial sideband modulation. Historically, in an ideal PAL I transmission, the lower vestigial sideband should begin to roll off at 1.25 MHz below the carrier frequency, falling to zero at 1.75 MHz below the video carrier. However, to allow sufficient room for the adjacent NICAM signals, VSB now tends to begin at 0.75 MHz and falls to zero at 1.25 MHz. Figure 6 shows the AM-VSB spectrum of a TV signal.

A typical cable TV system will carry between 40 and 45 AM-VSB channels, which with time sharing allows for the

carriage of, maybe, 50 TV programmes. These will be sourced from off-air terrestrial TV (BBC1 and 2, ITV, Channel 4 and Channel 5), satellite (BSkyB, CNN, etc) and local programming from tape. Irrespective of source, each programme is re-modulated from its original format into AM-VSB for transmission over the network. At the customer premises each channel can either be directly tuned on the TV set (cable ready) or selected by the analogue set-top box for display by the TV set.

Most cable operators provide tiered levels of programming such as basic and premium. Access to the premium channels is controlled by the customer's set-top box which is used to unencrypt these channels under control of the cable operator's conditional access system.

4.2 Digital television

The need for more capacity in satellite, cable and terrestrial networks to carry ever more TV channels has

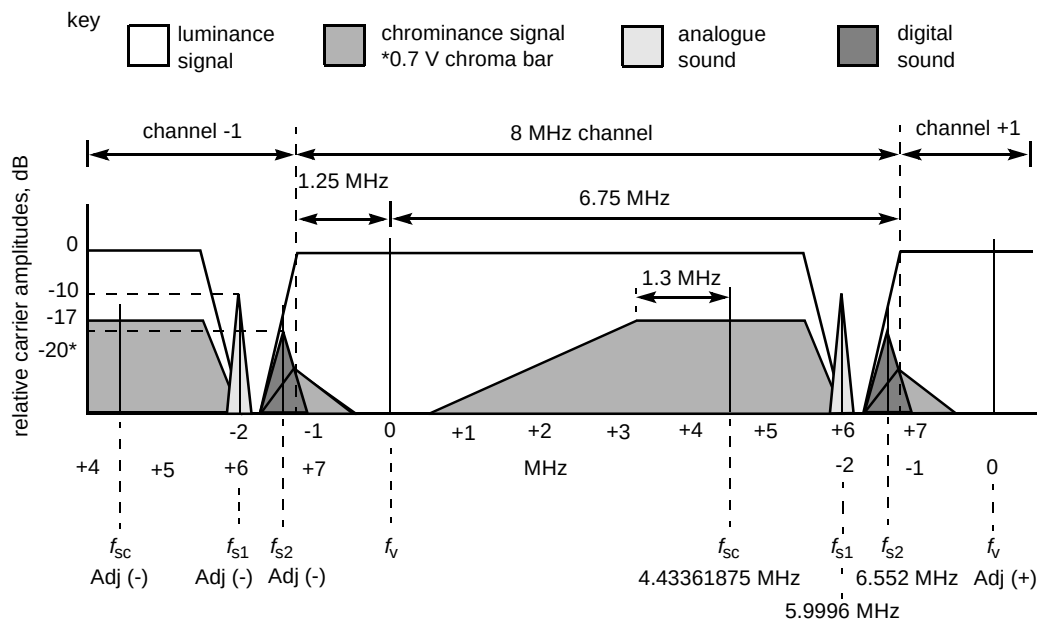


Fig 6 AM-VSB spectrum for a PAL-I TV signal.

been recognised. Digital transmission can provide the extra capacity needed. The standard chosen to provide digital TV is Digital Video Broadcasting (DVB). The benefit of an agreed digital TV standard helps both the vendors and cable operators, as this will eventually reduce cost through component availability and provide for agreed interfaces. In addition, the existence of a standard is often the trigger to entice major silicon suppliers to invest in large scale integration (LSI) resulting in a drastic reduction in size, together with the production of application-specific integrated circuits (ASICs) to help reduce costs.

Digital video broadcasting

The division of the world's television services between NTSC, PAL, SECAM and MAC built substantial barriers between continents and within Europe. The introduction of a Digital Television Standard has started to change this, as all systems will have the same basic standard throughout Europe. The Digital Video Broadcasting standard provides for a common digital compression scheme (MPEG) together with a common conditional access (CA) interface, services information (SI) and electronic programme guide (EPG). The core element of this standard has been designed around the MPEG-2 transport layer, but incorporates the subtle changes required to accommodate the differences between transmission over satellite, terrestrial or cable.

Digital video

ITU-R Recommendation 601 states that each of the luminance and colour difference signals Y, B-Y and R-Y are sampled in the following manner.

Luminance	Y	13.5 MHz using 10 bits
Blue - Luminance	B-Y	6.75 MHz using 10 bits
Red - Luminance	R-Y	6.75 MHz using 10 bits

This would result in a 10-bit parallel data rate of 27M/word or a serial data rate of 270 Mbit/s. Clearly some form of video compression and sophisticated modulation techniques will be required if these signals are to be transported within the 8 MHz analogue slots currently used in most European terrestrial and cable transmission. This compression technique is the subject of a family of standards and is commonly known as MPEG (moving pictures expert group) standard ISO/IEC 13818. The units that carry out this compression process are more commonly referred to as MPEG encoders.

The compressed digital TV signal is framed into a transport stream as shown in Fig 7.

The DVB transport stream can be transmitted over satellite, terrestrial or cable systems. The nature of each of these systems is different requiring the use of different modulation schemes to achieve best results.

Digital cable transmission

Taking into account the limitations of an 8 MHz wide TV channel and a roll-off factor of 0.15, the maximum theoretical symbol rate in the channel is 6.96 Mbaud.

Table 1 shows examples of useful bit rates and total bit rates for efficient use on cable networks [2].

The first row in Table 1 shows that 64 QAM can be used for the efficient transparent transmission of a digital satellite sourced 38.1 Mbit/s stream within the 8 MHz cable channel but at a total cable bit rate of 41.34 Mbit/s. A 31.672 Mbit/s PDH source can similarly be carried within the 8 MHz cable channel using 32 QAM modulation.

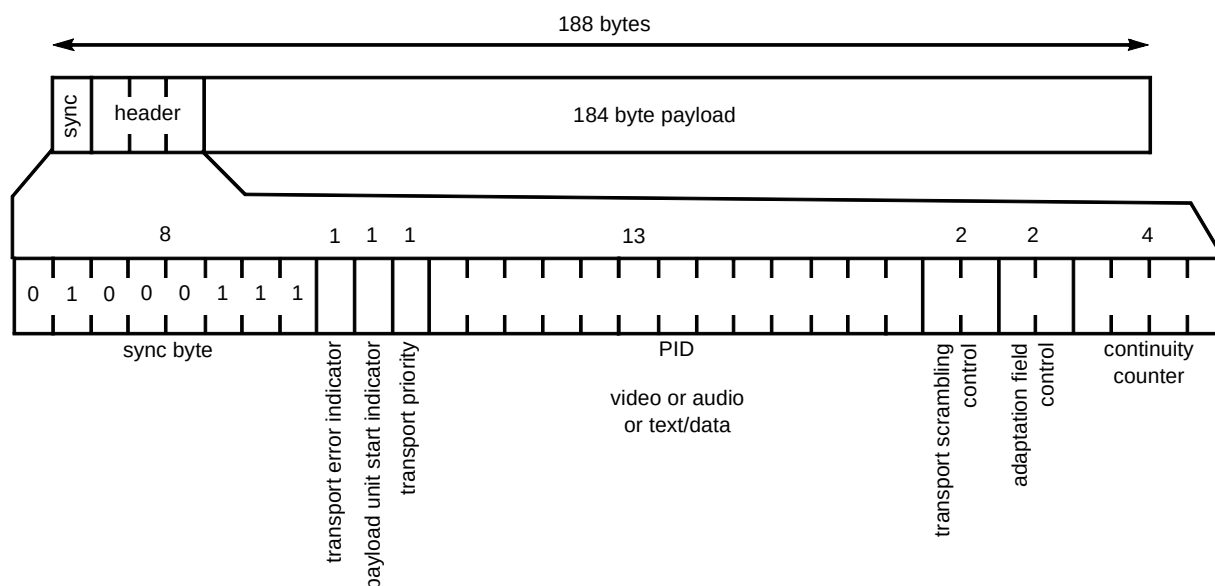


Fig 7 Structure of the DVB transport stream.

Table 1 Bit rates and total bit rates for efficient use on cable networks [2].

MPEG-2 transport layer useful bit rate (Mbit/s)	Total bit rate including Reed Solomon (204, 188) (Mbit/s)	Cable symbol rate (Mbaud)	Occupied bandwidth (MHz)	Modulation scheme
38.1	41.34	6.89	7.92	64 QAM
31.9	34.61	6.92	7.96	32 QAM
25.2	27.34	6.84	7.86	16 QAM
31.672 (PDH)	34.367	6.87	7.9	32 QAM
18.9	20.52	3.42	3.93	64 QAM
16	17.4	3.48	4	32 QAM
12.8	13.92	3.48	4	16 QAM
9.6	10.44	1.74	2	64 QAM
8	8.7	1.74	2	32 QAM
6.4	6.96	1.74	2	16 QAM

5. Digital services

Using the techniques described in the previous section, cable systems can be used to carry a variety of other digital services in addition to digital TV. These services include digital music such as Digital Music Express™ (DMX), games (Sega channel) and fast data access using cable data modems. This section will deal only with the last of these three services.

5.1 The cable modem

With the ever increasing interest in Internet browsing, e-mail and general data transfer, it has become apparent that cable TV networks have the potential to offer a much higher bandwidth than their PSTN counterpart. More recently technical advances in this area have facilitated cable modems to have the same physical appearance as a PSTN modem, making them available either in small, stand-alone units, or as cards to plug into a PC expansion slot. Interconnection between the cable modem and the cable network, is via co-axial cable in exactly the same way as the set-top unit.

The cable modem utilises unused TV channels in the forward direction, and the reverse path bandwidth of the cable network for the return.

5.2 Cable modem standards

Co-axial cable has been used as a data transmission medium for many years, but until recently this has been limited to campus size areas and data rates of typically up to 19.2 kbit/s. Cable networks specifically designed for data tended to be mid-split-band with equal spectrum available for both upstream and downstream to reflect the symmetrical nature of the data being transmitted. Large cable networks meant for TV distribution are asymmetric

by nature and therefore do not lend themselves well to traditional data transmission. However, the emergence of the Internet and Web browsing tends to demand greater data bandwidth downstream and more limited bandwidth upstream. As long as a return path is provided, cable systems seem to fit the requirements for Internet access very well.

Traditional cable data modems use protocols, such as extended Ethernet, which are inefficient when large numbers of users wish to access data simultaneously. The search for a standardised protocol which would allow thousands of users to access data bandwidth simultaneously led to the setting up of an IEEE group to co-ordinate activity on cable modem standards and to produce an open standard that would be accepted by the ITU-T. IEEE 802.14, as it is known, has attracted large numbers of manufacturers and operators with a vested interest in getting a suitable standard accepted and equipment made available at low cost.

Perceived delays achieving a standard protocol have led to several manufacturers developing their own proprietary protocols. Companies such as Motorola and Com21 have had considerable success selling their own modem systems to cable operators world-wide who are anxious to commence commercial data services as soon as they can. At the end of 1997 there were around 110 000 cable modems in service in the USA, all using proprietary equipment.

Frustration over continuing delays in IEEE802.14 being finalised led six of the largest North American cable companies to ask their joint research company, CableLabs, to develop a simpler data standard that would allow interoperability between various manufacturers' equipment. CableLabs' own cable platform would be used as an 'incubator' to test various manufacturers' data modem chip sets and, when fully assembled into working cable modems, the complete data modem system.

This activity is called the multimedia cable network system (MCNS). Neither CableLabs nor the cable companies are allowed to propose new standards to the official standards setting bodies, and therefore the Society of Cable Telecommunications Engineers (SCTE), which has that right, is also now part of the MCNS group.

A further development of MCNS has been to set up 'PacketCable' as a set of rules to allow effective use of cable modems for video streaming (MPEG), voice (IP) telephony, and all other multimedia applications. A further set of rules has been proposed called 'OpenCable' to encourage the major manufacturers of digital set-top boxes to produce interoperable designs which can incorporate a cable modem to allow connection to the TV set as the display device or to a PC for more conventional use.

5.3 The IEEE802.14 data standard

The charter for IEEE802.14 was to produce a single media access control (MAC) and multiple physical (PHY) specifications that would be compliant with IEEE802 as well as being both ATM and digital TV compatible.

Although IEEE802.14 is American based, they have worked closely with the European-based Digital Video Broadcast (DVB) group, DAVIC, ATM Forum, RBB and with MCNS to ensure that the standard can also be used outside North America.

The size of the task can be judged from the long delays that have been experienced in getting to the stage of releasing the standard. Work started on IEEE802.14 in 1995 and by July 1997 the group had released a draft standard for comment. This included the following physical layer recommendations for downstream:

- ITU-R Recommendation J.83 Annex B with the optional variable-depth interleaving and options to accommodate requirements by the SCTE Digital Video Standard (DVS) standards working group and MCNS:
 - frequency range 91—857 MHz,
 - level adjustable over the range +50 to +61 dBmV,
 - modulation type 64 or 256 QAM,
 - symbol rate 5.056941—5.19 Mbit/s,
- ITU-R recommendations J.83 Annex A and C with options or extensions to accommodate requirements of DAVIC/DVB,

- optional use of MPEG2-TS PID multiplexing;

and in the upstream recommendation:

- support for QPSK and QAM with frequency ranges from 5—42 MHz.

Work continues on developing advanced upstream specifications using trellis coding, multicarrier and spread spectrum to increase spectral efficiency. Further recommendations for forward error correction (FEC) have also been drafted.

In the MAC (media access control) layer acquisition, security class of service, quality of service and management have all been addressed and recommendations proposed to the group for comment.

The draft specification was submitted as an informal contribution to US Study Group B in August 1997. The approved draft for ballot is expected during the second half of 1998.

5.4 MCNS data standard

The Society of Cable Telecommunications Engineers (SCTE) Data Standards Subcommittee (DSS) adopted the MCNS Data Over Cable Radio Frequency Interface Specification as its standard (DSS-97-2) in July 1997 and submitted it to the US State Department Study Group 9 as a contribution to the ITU-T for recognition as an international standard.

This standard has the internal and external network interface of a system that allows transparent two-way transport of Internet protocol (IP) traffic over cable-based networks; it is referred to as DOCSIS (Data Over Cable System Interface Specification) and is becoming the *de facto* standard in the USA.

5.5 DVB/DAVIC data standard

EuroCable Labs, members of which include several European Cable operators and manufacturers, is backing the standard known as DVB/DAVIC. This is based on the DVB digital TV system being deployed in Europe.

The major difference between this and the DOCSIS standard is that the DVB/DAVIC standard uses fixed-length ATM cells for transport, rather than variable-length IP packets. The potential advantage of that is that it is better suited to the transport of delay-sensitive services such as voice and videotelephony. At the time of writing, it is not clear which standard will be adopted in Europe, but at least two of the major UK cable companies are backing DOCSIS.

5.6 Modulation techniques

Early cable modems used simple modulation schemes like amplitude shift keying (ASK) and frequency shift keying (FSK). At high data rates the amount of bandwidth occupied by the modulated signal becomes significant. Multilevel modulation schemes like 16-VSB and 64 QAM are much more spectrally efficient although they do require a much higher CNR, than ASK or FSK, to operate with low error rates.

Spectral efficiency is usually rated according to the number of bits of data that can be 'compressed' into each hertz of bandwidth. Quadrature amplitude modulation with 64 levels theoretically gives 6 bits/Hz, although in practice little more than 5 bits/Hz can be achieved due to implementation limitations. This means that a 30 Mbit/s data stream will occupy about 6 MHz of spectrum and 40 Mbit/s will occupy about 8 MHz. In North American TV, channel spacing is 6 MHz and in Europe it is 8 MHz, so these data rates provide a good fit to the cable system 'raster'. It is no coincidence that digital TV will also use 64 QAM modulation

Figure 8 shows the constellation of a 64 QAM modulated data signal.

64 QAM modulation requires a theoretical CNR of around 28.6 dB for an error rate of 10^{-9} [3] and this can easily be achieved in a cable system where the CNR has been designed to be in excess of 40 dB to give good TV picture quality. The excess CNR can be used either to reduce the transmit level of digital channels (a -10dB reduction has been proposed by some equipment suppliers) and hence increase the cable loading, or to make use of the higher frequency ranges where drop cable losses are high and hence give low signal level at the customer premises.

In the upstream direction the available CNR will inevitably be lower due to a combination of 'funnelled noise' from multiple sources and low-cost, low-performance upstream laser transmitters and receivers. Noise ingress is also likely to be higher and so a more 'rugged' modulation scheme is required at the expense of poorer spectral efficiency. Quadrature phase shift keying (QPSK) is the most common choice, providing a theoretical spectral efficiency of 2 bits/Hz. In practice, this is more likely to be around 1.5 bits/Hz, again due to implementation limitations. The limited upstream bandwidth and low spectral efficiency of the modulation scheme mean that upstream data rates are limited to about 10 Mbit/s, shared across all the customers on each optical node of the network. It should be noted that the 10 Mbit/s can be shared by between as few as 50 and as many as 2500 customers, depending on node architecture.

Alternative modulation schemes have been proposed for both downstream and upstream, with 256 QAM downstream and 16 QAM upstream. This is already being seriously considered for the next generation of digital TV and data modems.

6. Cable telephony

6.1 Integrated and overlay

Until recently cable TV and telephony have been very different services even though they often served the same customer premises. Economic advantages are to be had in providing both services together — billing and service management functions can be shared and costs of account management reduced. In their search for efficiency, telephony companies have been looking at techniques to provide broadband services over twisted pairs using techniques such as ADSL. In their turn, cable companies are looking at ways to provide telephony to their customers.

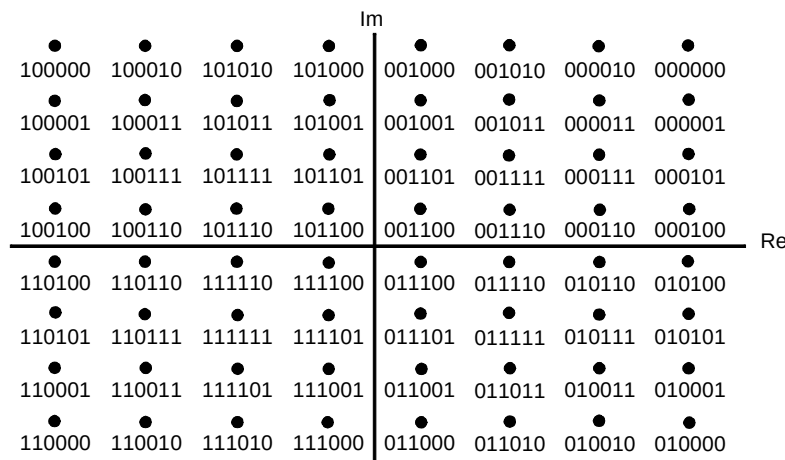


Fig 8 The 64 levels of the 64 quadrature amplitude modulated (64 QAM) carrier signal — 64 possible phase/amplitude symbols allow six bits per symbol.

The choice is between transmitting the telephony signals using the existing coaxial cable network, which is called integrated telephony, or of setting up a separate overlay telephony network using standard twisted pair technology which shares the same routes as the coaxial network. Overlay was the most attractive option for many UK cable companies as they were building their networks from scratch in the early 1990s — the marginal cost of putting in a twisted pair network alongside their coaxial cable network was low. For operators with existing cable networks the integrated solution is likely to be far more attractive since it avoids expensive civil works.

6.2 The technology

Cable systems have been optimised to give a low-noise, highly linear forward transmission path. These characteristics are set by the requirements of the AM-VSB television signal, resulting in an SNR of better than 43 dB in an 8 MHz channel bandwidth (6 MHz in the USA). A number of telephony channels can be aggregated together and sent along a single 8 or 6 MHz downstream channel. For example, one well-known equipment manufacturer's system sends 360 simultaneous 64 kbit/s calls down a single 6 MHz forward channel, using 64 QAM modulation.

6.3 Return path

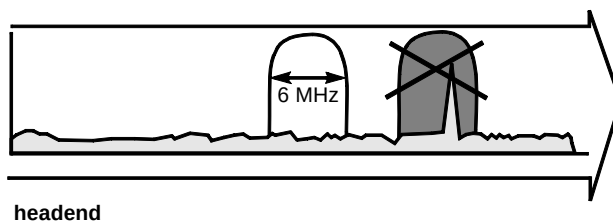
Getting the voice signal back to the headend is where the challenge lies in integrated telephony. The return path of an HFC system is achieved by splitting the frequency spectrum into two, typically using the range 5—50 MHz for the upstream direction, and 70 MHz and above for the forward direction.

In the reverse direction all the taps and splitters which divided the TV signal act as combiners, since the signal direction is reversed, and this means that all the reverse path signals from all the customers are combined. This funnels in all the noise and interference from all customer locations as well as the wanted signal, which makes a much more noisy channel than the forward path. To limit the amount of noise, the return path is typically split up as shown in Fig 9. In this way the signal-to-noise ratio does not degrade too much, and the network is given some protection against a single interference source taking all customer return paths out of action.

6.4 Solutions

Reliable operation in the return path needs a modulation scheme which can operate well in the noisy return path, combined with frequency agility to move away from channels which are jammed by single-frequency interference. For example, one manufacturer uses orthogonal frequency division multiplexing (OFDM) in the return path. OFDM divides the available frequency spectrum into many narrow bands, each of which can be individually modulated.

Downstream - interference narrower than the channel width knocks out the whole channel and all information is lost. This is acceptable since such interference is not present in the downstream channel.



OFDM upstream - interference narrower than the channel width knocks out only part of the information but most data remains intact.

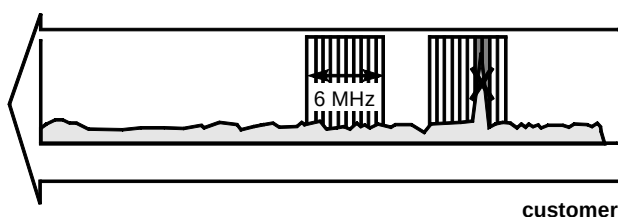


Fig 9 The effect of narrowband interference on upstream and downstream signals.

With a conventional modulation system, narrowband interference in any channel would knock out the entire channel's capacity. By splitting the return path into hundreds of channels, the information can be moved away from the few narrowband channels affected by interference to unaffected channels. This process can be controlled by monitoring the bit error rate received at the headend.

The headend control equipment (HDT) can move a 'jammed' channel to a clear one by controlling the customer unit via the clean downstream channel.

6.5 Concentration over the HFC

In a typical residential service, about 10:1 concentration can be achieved by having the customer unit request service from the HDT on demand. Dynamic allocation of channels is already a part of the mechanism to deal with interference and the HDT therefore instructs the customer unit to go to the next free return path frequency on demand.

6.6 Voice and data

Cable data modems and cable telephony systems have many parts in common and there is increasing interest in integrated voice and telephony products. The West End Westbound 9600 was the first combined voice/data product, but other manufacturers have looked at integrating data with voice.

Though hampered by a current lack of open standards, a single platform to offer advanced data and voice services via their co-axial networks is a very attractive proposition for cable operators.

7. Cable futures

Although cable companies have historically not been very proactive in exploiting the potential of their broadband networks, the opportunities offered by delivering more TV channels and the possibilities of very high speed Internet access have not been missed. UK cable companies are currently planning to introduce digital TV before the year 2000, to provide both the capacity for carrying the expected increase in satellite-sourced digital TV channels, and also new VOD services such as staggered-start single-film channels (NVOD).

Recent interest in cable modems has reduced but is expected to rise again with the introduction of combined digital set-top boxes and cable modem in a single unit.

8. Conclusions

Cable technology is able to fulfil the requirements of local access, in particular the efficient and flexible delivery of broadband services such as digital TV. Ongoing development of standards for cable modems will allow cable operators to deliver fast data services to cable customers in their franchise areas, effectively forming cable-based wide area networks (WANs), which in turn will be linked to the cable company high-speed backbone networks. These backbone networks will carry digital TV, data and voice services from a central headend to as many as 50 remote headends corresponding to the existing cable headends in each franchise.

As demand for bandwidth grows, the fibre portion of HFC will drive deeper into the network until it reaches individual homes and the 'fibre to the home' (FTTH) vision will be achieved.

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Sam Jewell joined BT in 1966 as an apprentice in Reading. From 1974 to 1984 he taught microwave, line transmission, television switching and digital techniques at BTTC Stone. In 1984 he moved to Martlesham Heath to work on the Westminster Switched Star Network where he was responsible for video performance measurements over the optical primary links. Subsequently he was responsible for the design of the broadband optical transmission system used in the Bishops Stortford BIDs trial. He is currently Product Engineering Manager, Cable TV at BTL.



Jeff Patmore joined BT as an apprentice in 1970. He worked on the maintenance and commissioning of microwave radio and coaxial line systems within the Telecom Tower complex, moving into an operations management role in 1982.

In 1989 he joined the BT Cable Television subsidiary business, with responsibility for network and computer systems maintenance and build.

In 1996 he moved to a strategic role working with teams at BT Laboratories managing the upgrade of the Westminster and Milton Keynes broadband networks, with the objective of increasing both reliability and services offered. Recently he has been investigating new services for deployment on BT Cable TV networks.



Kevin Stalley joined BT as an apprentice in 1971, initially working on transmission and maintenance both for radio and cable-based systems, utilising analogue and digital transmission techniques. He transferred to Martlesham Heath in 1978 to work on submarine cable systems, before moving on to radio, fibre systems and finally to multimedia platforms. Recent work has involved cable TV systems and trials including design work on the following systems: BT Switched Star Network in London, the Bishops Stortford fibre trial (fibre to the home) using BPON and TPON topologies, and more recently HFC systems for BT's Cable Television Services, including technical consultancy and system design. His current area of expertise has been to exploit the recent advances in Digital Video Broadcasting for cable systems and to maximise this impact on cable as an alternative access network.



Richard Mudhur graduated from Imperial College, London in 1981. He worked as a broadcast engineer for BBC Television, moving on to design broadcast test and digital effects equipment. After taking an MSc in 1988 he joined BT Laboratories working on the LLOFT, an advanced cable TV trial using optical fibre transmission to the home.

Over the last few years he has worked on several innovative cable transmission systems and has headed a RACE project developing cable telephony solutions. He is currently working on cable telephony and advanced digital applications.